

NOVOSIBIRSK TOLMACHEVO, Russian Federation

WMO: 296340

Lat: 55.013N

Lon: 82.651E

Elev: 111

StdP: 100.00

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.2	-33.1	-40.1	0.1	-36.2	-36.9	0.1	-33.2	11.7	-6.9	10.1	-7.8	1.4	20	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.8	18.9	28.0	18.2	26.3	17.5	20.7	26.7	19.8	25.5	18.9	24.4	4.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.9	23.1	17.9	13.0	22.3	16.9	12.2	21.7	60.5	26.7	57.5	25.4	54.3	24.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.1	8.6	7.6	-38.6	32.6	4.2	2.0	-41.6	34.0	-44.0	35.2	-46.4	36.3	-49.4	37.7	
			-38.6	22.6	4.1	1.2	-41.5	23.4	-43.9	24.1	-46.2	24.8	-49.2	25.6	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-17.9	-14.5	-6.7	3.8	11.7	17.2	19.0	16.6	10.3	3.1	-7.0	-14.0
	DBStd	14.28	8.60	7.80	6.60	5.53	5.50	4.18	3.19	3.66	4.59	5.13	7.99	8.46
	HDD10.0	3846	866	686	519	198	48	3	0	1	52	219	508	743
	HDD18.3	6122	1124	919	778	437	214	68	31	76	243	472	758	1002
	CDD10.0	886	0	0	0	12	99	220	281	207	61	6	0	0
	CDD18.3	121	0	0	0	0	7	35	53	23	3	0	0	0
	CDH23.3	1312	0	0	0	3	142	391	476	257	43	0	0	0
	CDH26.7	316	0	0	0	0	34	94	113	64	9	0	0	0
Wind	WSAvg	3.6	3.5	3.6	3.9	4.1	4.1	3.2	2.8	2.9	3.3	3.8	4.0	4.0
Precipitation	PrecAvg	471	24	19	21	26	35	57	70	58	41	44	40	36
	PrecMax	658	47	38	43	49	74	122	111	144	92	82	69	61
	PrecMin	325	5	2	4	7	2	19	10	9	15	14	23	16
	PrecStd	83	11	9	10	12	20	28	30	28	19	17	14	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	1.9	7.5	22.9	30.2	31.1	31.9	31.1	28.0	20.2	8.9	2.0
		MCWB	0.1	0.5	3.5	12.2	16.0	18.5	20.1	19.6	15.5	11.6	6.5	0.7
	2%	DB	-1.8	-0.1	5.1	19.0	26.9	29.1	29.3	28.2	23.9	16.8	6.2	0.8
		MCWB	-2.6	-1.2	2.1	10.4	14.6	18.2	19.6	18.8	14.5	10.0	4.3	0.1
	5%	DB	-3.5	-1.8	3.2	15.9	24.6	27.4	27.9	26.1	21.0	13.2	4.2	-1.0
		MCWB	-4.3	-2.8	1.1	8.8	14.0	17.9	19.2	17.8	13.5	8.3	2.6	-1.9
	10%	DB	-5.8	-3.8	2.0	12.9	21.9	25.8	26.1	24.1	18.2	10.2	2.2	-3.1
		MCWB	-6.7	-4.7	0.2	7.5	12.9	17.3	18.5	16.9	12.1	7.0	1.0	-4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.5	4.1	13.2	17.8	21.1	22.4	21.3	17.2	12.2	7.2	0.7
		MCDB	0.8	1.3	6.8	19.4	27.4	27.5	27.8	28.0	25.0	18.2	8.3	1.2
	2%	WB	-2.4	-1.2	2.5	11.0	16.1	20.0	21.2	19.8	15.3	10.5	4.4	-0.2
		MCDB	-1.7	-0.3	4.4	17.0	23.9	26.6	26.8	26.3	21.6	15.2	5.8	0.4
	5%	WB	-4.3	-2.8	1.4	9.2	14.7	19.0	20.2	18.8	14.2	8.9	2.7	-2.0
		MCDB	-3.5	-2.0	3.1	15.6	22.2	25.1	25.5	24.4	20.1	12.8	4.0	-1.1
	10%	WB	-6.4	-4.5	0.2	7.5	13.4	18.0	19.5	17.7	12.9	7.3	1.1	-4.0
		MCDB	-5.6	-3.6	1.8	12.3	20.6	23.8	24.5	22.5	17.3	10.2	2.1	-3.2
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.0	9.5	10.1	12.2	11.3	10.8	11.3	10.6	8.3	6.7	7.5
		MCDBR	7.9	8.1	9.1	15.1	16.0	14.6	13.5	14.7	15.1	13.6	6.6	7.1
		MCWBR	7.8	7.6	7.0	8.3	7.2	6.3	5.8	6.6	7.3	8.0	5.3	7.0
	5% WB	MCDBR	7.9	7.7	7.7	13.9	14.1	12.4	11.4	12.7	13.7	12.1	6.5	7.2
		MCWBR	7.8	7.5	6.3	8.4	7.4	6.1	5.7	6.4	7.0	7.7	5.5	7.2
Clear-Sky Solar Irradiance	taub	0.294	0.328	0.378	0.410	0.407	0.410	0.439	0.413	0.373	0.355	0.327	0.297	
	taud	2.005	1.960	1.912	2.085	2.213	2.288	2.207	2.261	2.320	2.235	2.046	1.954	
	Ebn at Noon	587	706	760	805	835	836	801	798	782	688	537	473	
	Edh at Noon	71	108	144	140	131	123	131	117	97	83	69	60	
All-Sky Solar Radiation	RadAvg	0.94	1.97	3.44	4.64	5.34	6.01	5.38	4.54	3.06	1.54	0.75	0.60	
	RadStd	0.08	0.11	0.25	0.29	0.58	0.54	0.35	0.34	0.36	0.20	0.13	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page